

Investigation 23: Identify the unknowns I

The situation

A freak storm has forced rainwater to flood down and through the chemical storage room. Labels have smudged and some have completely fallen off a number of bottles. Luckily you know that one set of the now unlabelled bottles contains solutions of cations and a second set contains solutions of anions.

The task

Design and carry out an investigation that will identify the set of unlabelled solutions of different cations with the same anion (nitrate ion) and a second set of solutions of different anions with the same cation (sodium ion).

Identification of cations

Set 1 (unknown cations): 0.1 mol L⁻¹ nitrate solutions of iron(II), lead, silver, copper(II) and magnesium, randomly labelled A, B, C, D and E.

Set A (known test solutions): 0.1 mol L⁻¹ solutions of sodium hydroxide, ammonia, sodium chloride and potassium iodide.

Identification of anions

Set 2 (unknown anions): 0.1 mol L⁻¹ solutions of sodium carbonate, sodium hydroxide, sodium chloride, sodium sulfate or sodium nitrate randomly labeled as unknown solution F, G, H, I and J.

Set B (known test solutions): 0.1 mol L⁻¹ solutions of lead nitrate, silver nitrate, iron(II) sulfate (fresh), copper(II) sulfate and sulfuric acid.

Planning the investigation

1. Plan an investigation to correctly identify the unknown cations and anions.
2. Write out your proposed plan, list the equipment required and identify the safety requirements.

SAFETY NOTE:

- Check your plan with your teacher before you start.
- Sulfuric acid is corrosive! Handle with care.

Conducting the investigation

1. Conduct the investigation, collecting and recording the data as you proceed.

Processing the data

1. From your observations identify each of the unknowns in the set. Include the reasons for your identifications and write chemical equations for all reactions that were observed.
2. Write an ionic equation showing how each precipitate forms.

Evaluating the investigation

1. Evaluate the effectiveness of your procedure and describe any modifications you would make to improve it.
2. Discuss your confidence in the findings of the investigation.

Notes

Use precipitation reactions to determine which cation you have in each bottle. You must make a record of each experimental test you do and explain how/why your testing determines the correct cation.

Investigation 24: Identify the unknowns II

Notes

Different substances dissolve in water to varying extents. The different solubilities of these substances have several applications. For example, chemists use knowledge of solubilities to identify the presence of certain ions, to prepare compounds, in electrolytic processes, horticulture and in rehydration. It also helps in various forensic tests and separation methods.

The task

Apply the solubility rules to identify a set of unknown solids or solutions.

Your teacher will allocate you one or more of the following sets of solids or solutions. Each member of the set provided will be labeled only as A, B, C ... etc.

Set 1: 1 mol L⁻¹ solutions of BaCl₂, CuSO₄, H₂SO₄, NaCl randomly labeled as A, B, C, D.

Set 2: 1 mol L⁻¹ solutions of AgNO₃, ZnSO₄, BaCl₂, NaCl randomly labeled as G, H, I, J.

Set 3: 1 mol L⁻¹ solutions of H₂SO₄, BaCl₂, Na₂CO₃, Pb(NO₃)₂, KOH and 2 mol L⁻¹ HCl randomly labeled as M, N, O, P, Q, R.

Set 4: Solid samples of BaCl₂, Zn(NO₃)₂, Na₂CO₃, CaCO₃ randomly labeled as W, X, Y, Z. (Distilled water, dilute hydrochloric acid and limewater solution are also provided to test the unknown solid samples.)

It will be necessary for you to plan a procedure to identify the unknowns provided, carry out the procedure and evaluate your results.

Equipment

Unknown samples in Set 1, 2, 3 or 4 as listed above with each chemical labeled only as A, B, C etc.

set 1-3 solutions in dropper bottles

set 4 solid samples in small sample bottles; if Set 4 unknowns are used, the following solutions in labelled dropper bottles as well as delivery tubes are also provided:

hydrochloric acid [HCl] 1 mol L⁻¹

limewater solution [Ca(OH)₂]

spatula

test tube racks

test tubes (several)

distilled water

beakers (two 100 mL)

Planning the investigation

1. Plan an investigation to identify the unknown solids or solutions provided. For Sets 1-3 you may only use the solutions provided. For Set 4 the solid samples can be tested with the hydrochloric acid, limewater and distilled water provided. For each test, use a minimum amount of sample - for solids about a pea-sized sample; for solutions about 2-3 mL.
2. Write out your proposed plan, list the equipment required and identify the safety requirements.
3. Check the proposed plan with your teacher.

SAFETY NOTE:

- Check your plan with your teacher before you commence.

Conducting the investigation

Conduct the investigation, collecting and recording the data in a table as you proceed.

Processing the data

From your observations identify each of the unknowns in the set. Include the reasons for your identifications and write chemical equations for all reactions that were observed.

Evaluating the investigation

1. Evaluate the effectiveness of your procedure and describe any modifications you would make to improve it.
2. Discuss your confidence in the findings of the investigation.

Notes